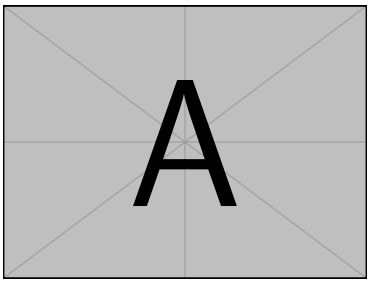

LaTeX Complete Reference

A Comprehensive Learning Guide

Your Name

April 1, 2026

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Chapter 1

Document Structure & Document Classes

Every LaTeX document consists of two main parts:

1. **Preamble** — everything before `\begin{document}`
2. **Body** — the content between `\begin{document}` and `\end{document}`

1.1 Document Classes

The `\documentclass` command must be the very first command. It defines the type of document.

Class	Purpose	Sectioning
`article`	Short documents, journal papers	section, subsection
`report`	Longer documents, theses	chapter, section
`book`	Full books	part, chapter, section
`letter`	Letters	—
`beamer`	Presentations/slides	frame
`scrartcl`	KOMA-Script article	section
`scrreprt`	KOMA-Script report	chapter
`scrbook`	KOMA-Script book	part, chapter

Table 1.1: Common document classes in LaTeX

1.1.1 Document Class Options

Class options go in square brackets: `\documentclass[options]{class}`

Option	Effect
`12pt`	Sets font size (10pt, 11pt, 12pt)
`a4paper`	Sets paper size (a4paper, letterpaper, legalpaper)
`twoside`	Formats for two-sided printing
`oneside`	Formats for one-sided printing (default)
`landscape`	Landscape orientation
`draft`	Draft mode — marks overfull boxes
`final`	Final mode (default)
`twocolumn`	Two-column layout
`titlepage`	Title on a separate page
`openright`	Chapters start on right-hand page
`leqno`	Equation numbers on left
`fleqn`	Left-aligned equations

Table 1.2: Common document class options

```latex
\documentclass[12pt, a4paper]{article}

% === PREAMBLE ===
% Load packages, define commands, set options here
\usepackage[utf8]{inputenc}
\usepackage{graphicx}
\title{My First Document}
\author{Author Name}

% === BODY ===
\begin{document}
\maketitle
Hello, World!
\end{document}
```

Listing 1.1: Basic document structure

1.2 Sectioning Commands

LaTeX provides a hierarchy of sectioning commands:

```latex
\part{Part Title}           % Level -1 (book/report only)
\chapter{Chapter Title}     % Level 0  (book/report only)
\section{Section Title}     % Level 1
\subsection{Subsection}     % Level 2
\subsubsection{Sub-sub}     % Level 3
\paragraph{Paragraph}       % Level 4
\subparagraph{Subparagraph}% Level 5
```

Listing 1.2: Sectioning hierarchy

1.2.1 Starred Versions

Each sectioning command has a starred version (e.g., `\section*{Title}`) that:

- Does *not* add a number
- Does *not* appear in the table of contents

1.3 Input and Include

Split large documents into multiple files:

```
% \input{file}      -- inserts file content inline (no page break)
% \include{file}    -- inserts file with \clearpage before and after
%                      (cannot be nested)

\input{chapters/chapter1}      % Preferred for small files
\include{chapters/chapter2}    % Preferred for chapters
```

Listing 1.3: File inclusion commands

Feature	`\input`	`\include`
Page breaks	No	Yes
Can be nested	Yes	No
Selective compilation	No	Yes (`\includeonly`)

Table 1.3: Comparison of `\input` vs `\include`

1.4 Comments

```
% This is a single-line comment.
% Everything after % on the line is ignored.

% For multi-line comments, use the verbatim package:
\usepackage{verbatim}
\begin{comment}
This entire block is commented out.
Line two of the comment.
Line three.
\end{comment}
```

Listing 1.4: Comments in LaTeX

1.5 Special Characters

The following characters are **reserved** in LaTeX and need special handling:

Character	**Purpose**	**To Print It**
`#`	Macro parameter	`\#`
`$`	Math mode	`\$`
`%`	Comment	`\%`
`&`	Table column sep.	`\&`
`_`	Subscript (math)	`\_`
`{`	Group begin	`\{`
`}`	Group end	`\}`
`~`	Non-breaking space	`\~{}` or `\textasciitilde`
`^`	Circumflex accent	`\^{}` or `\textasciicircum`
`\`	Command prefix	`\textbackslash`

Table 1.4: Special characters in LaTeX

Note: Accented characters can be typed directly in UTF-8: é, ü, ñ, ç, etc.

Chapter 2

Text Formatting & Fonts

2.1 Font Styles

LaTeX provides command pairs for font switching — one with *declaration syntax* (affects all following text) and one with *command syntax* (takes an argument).

Style	Declaration	Command
Bold	`\bfseries`	`\textbf{...}`
Italic	`\itshape`	`\textit{...}`
Slanted	`\slshape`	`\textsl{...}`
Monospace	`\ttfamily`	`\texttt{...}`
Sans-serif	`\sffamily`	`\textsf{...}`
Small Caps	`\scshape`	`\textsc{...}`
Medium	`\mdseries`	`\textmd{...}`
Upright	`\upshape`	`\textup{...}`

Table 2.1: Font style commands

2.1.1 Examples

- **Bold text** with `\textbf{Bold text}`
- *Italic text* with `\textit{Italic text}`
- Typewriter text with `\texttt{Typewriter text}`
- Sans-serif text with `\textsf{Sans-serif text}`
- Small Capitals with `\textsc{Small Capitals}`
- *Slanted text* with `\textsl{Slanted text}`
- *Emphasized text* — `\emph{}` toggles italic inside roman and roman inside italic
- *This is* nested *emphasis* — `\emph{This is \emph{nested} emphasis}`

2.2 Font Sizes

Size commands are *declarations* — they affect everything until the end of the group:

Command	Result
`\tiny`	Sample text
`\scriptsize`	Sample text
`\footnotesize`	Sample text
`\small`	Sample text
`\normalsize`	Sample text
`\large`	Sample text
`\Large`	Sample text
`\LARGE`	Sample text
`\huge`	Sample text
`\Huge`	Sample text

Table 2.2: Font size commands (relative to base document font size)

```
{\large This text is large.}
{\Large\bfseries This text is large and bold.}
```

Listing 2.1: Font size usage

2.3 Text Alignment

2.3.1 Centering

```
% Environment form:
\begin{center}
    This text is centered.
\end{center}

% Declaration form (affects rest of group):
{\centering This text is centered.\par}
```

Listing 2.2: Centering text

This is centered text using the `center` environment.

2.3.2 Flush Left (Ragged Right)

```
\begin{flushleft}
    This text is left-aligned.
\end{flushleft}
```

Listing 2.3: Left-aligned text

2.3.3 Flush Right (Ragged Left)

```latex
\begin{flushright}
    This text is right-aligned.
\end{flushright}
```

Listing 2.4: Right-aligned text

This is right-aligned text.

2.4 Line & Paragraph Spacing

```latex
% Single spacing (default)
% Double spacing:
\usepackage{setspace}
\doublespacing

% 1.5 spacing:
\onehalfspacing

% Custom spacing:
\setstretch{1.25}

% Line break (new line, same paragraph):
Line one.\\
Line two.

% Line break with extra space:
Line one.\\[12pt]
Line two with 12pt gap above.

% New paragraph (blank line in source):
Paragraph one.

Paragraph two.

% Horizontal spacing:
Word\hspace{1cm}Word          % 1cm horizontal space
Word\hspace{\fill}Word        % Stretchy space (pushes to margins)
Word\hfill Word               % Shorthand for \hspace{\fill}
```

Listing 2.5: Line spacing

2.4.1 Vertical Spacing

```latex
\vspace{1cm}          % 1cm vertical space
\vspace*{1cm}         % Forced space (even at page top)
\vspace{\fill}        % Stretchy vertical space
\vfill                % Shorthand for \vspace{\fill}
\bigskip              % Large vertical skip (~12pt)
\medskip              % Medium vertical skip (~6pt)
```

```
\smallskip              % Small vertical skip (~3pt)
```

Listing 2.6: Vertical spacing

2.4.2 Horizontal Spacing

Command	Width	Description
`\,` or `\thinspace`	∼3/18 quad	Thin space
`\:` or `\medspace`	∼4/18 quad	Medium space
`\;` or `\thickspace`	∼5/18 quad	Thick space
`\!`	−3/18 quad	Negative thin space
`\enspace`	0.5em	En-space
`\quad`	1em	Quad space
`\qquad`	2em	Double quad
`\~{}`	variable	Non-breaking space
`\hspace{length}`	variable	Custom horizontal space

Table 2.3: Horizontal spacing commands

2.5 Text Decoration & Color

2.5.1 Underline and Overline

```
\underline{Underlined text}
\overline{Overlined text}         % Works in math mode too
```

Listing 2.7: Underline and overline

Underlined text example.

2.5.2 Strikethrough (requires ulem package)

```
\usepackage[normalem]{ulem}

\sout{Strikethrough text}
\uwave{Wavy underline}
\xout{Crossed out with /}
```

Listing 2.8: Strikethrough

2.5.3 Text Color (xcolor package)

```
\textcolor{red}{Red text}
\textcolor{blue}{Blue text}
\textcolor{green!50!black}{Dark green text}

% Color as declaration:
{\color{red} This entire block is red.}
```

```latex

% Define custom colors:
\definecolor{mycolor}{RGB}{102, 178, 255}
\textcolor{mycolor}{Custom colored text}

% Background color (requires xcolor with table option):
\colorbox{yellow}{Highlighted text}
\fcolorbox{red}{yellow}{Box with red border and yellow background}
```

Listing 2.9: Using colors

Examples: Red, Blue, Dark Green, Highlighted

2.6 Accents and Special Characters

Accent	Command	Result	Alt (UTF-8)
Grave	`\‘{o}`	ò	ò
Acute	`\’{o}`	ó	ó
Circumflex	`\^{o}`	ô	ô
Tilde	`\~{o}`	õ	õ
Umlaut	`\"{o}`	ö	ö
Cedilla	`\c{c}`	ç	ç
Ogonek	`\k{a}`	ą	
Macron	`\={o}`	ō	
Breve	`\u{o}`	ŏ	
Dot	`\.{o}`	ȯ	
Ring	`\r{a}`	å	
Hacek	`\v{o}`	ǒ	
Double acute	`\H{o}`	ő	

Table 2.4: Accent commands

2.7 Dashes and Quotes

Type	Source	Result	Usage
Hyphen	`-`	mother-in-law	Compound words
En-dash	`--`	pages 1–10	Ranges
Em-dash	`---`	Yes—indeed	Interruptions
Minus	`$-$`	−	Mathematical minus

Table 2.5: Types of dashes

Type	**Source**	**Result**
Opening double quotes	` `	“Hello”
Closing double quotes	' '	“Hello”
Opening single quote	`	‘Hello’
Closing single quote	'	‘Hello’

Table 2.6: Quotation marks

Note: LaTeX does not use regular keyboard quotes ("). Always use backticks for opening quotes.

Chapter 3

Lists

3.1 Itemize (Bulleted Lists)

```
\begin{itemize}
    \item First item
    \item Second item
    \item Third item
\end{itemize}
```

Listing 3.1: Basic itemize list

Result:

- First item
- Second item
- Third item

3.1.1 Nested Itemize

```
\begin{itemize}
    \item First level
    \begin{itemize}
        \item Second level
        \begin{itemize}
            \item Third level
            \begin{itemize}
                \item Fourth level
            \end{itemize}
        \end{itemize}
    \end{itemize}
\end{itemize}
```

Listing 3.2: Nested itemize

Result:

- First level

- Second level
  - Third level
    - Fourth level

3.2 Enumerate (Numbered Lists)

```
\begin{enumerate}
    \item First item
    \item Second item
    \item Third item
\end{enumerate}
```

Listing 3.3: Basic enumerate list

Result:

1. First item
2. Second item
3. Third item

3.2.1 Nested Enumerate

1. First level
   (a) Second level
      i. Third level
         A. Fourth level
2. Back to first level

3.3 Description Lists

```
\begin{description}
    \item[Term 1] Definition or description of term 1.
    \item[Term 2] Definition or description of term 2.
    \item[Term 3] Definition or description of term 3.
\end{description}
```

Listing 3.4: Description list

Result:

LaTeX A document preparation system.

TeX The typesetting engine by Donald Knuth.

PDF Portable Document Format.

3.4 Customizing Lists with enumitem

The `enumitem` package provides powerful list customization.

3.4.1 Custom Labels

```latex
% Roman numerals:
\begin{enumerate}[label=\Roman*.]
    \item First
    \item Second
\end{enumerate}

% Alphabetical:
\begin{enumerate}[label=\alph*)]
    \item First
    \item Second
\end{enumerate}

% Custom format:
\begin{enumerate}[label=Step \arabic*:)]
    \item First
    \item Second
\end{enumerate}
```

Listing 3.5: Custom enumerate labels

Roman numerals:

I. First

II. Second

III. Third

Alphabetical:

a) First

b) Second

c) Third

Custom format:

Step 1: Gather requirements

Step 2: Design solution

Step 3: Implement

Step 4: Test

3.4.2 Custom Bullet Symbols

```latex
\begin{itemize}[label=$\star$]
    \item Star bullet
```

```latex
    \item Another star
\end{itemize}

\begin{itemize}[label=\textbullet]
    \item Standard bullet
\end{itemize}

\begin{itemize}[label=$\checkmark$]
    \item Checkmark bullet
\end{itemize}
```

Listing 3.6: Custom bullets

- ⋆ Star bullet one
- ⋆ Star bullet two
- ✓ Completed task one
- ✓ Completed task two

3.4.3 Spacing Control

```latex
\begin{itemize}[nosep]            % No extra spacing between items
    \item Compact item 1
    \item Compact item 2
\end{itemize}

\begin{itemize}[topsep=0pt]      % No space above/below list
    \item Item 1
    \item Item 2
\end{itemize}

\begin{enumerate}[start=5]       % Start counting from 5
    \item Fifth
    \item Sixth
\end{enumerate}
```

Listing 3.7: List spacing

Compact list (`nosep`):

- Compact item 1
- Compact item 2
- Compact item 3

Starting from 5:

5. Fifth
6. Sixth
7. Seventh

3.5 Inline Lists

```latex
\usepackage[inline]{enumitem}

The colors are:
\begin{enumerate*}[label=\arabic*.]
    \item red, \item green, and \item blue.
\end{enumerate*}
```

Listing 3.8: Inline list with enumitem

Note: Inline lists require `\usepackage[inline]{enumitem}`.

3.6 List Summary

Environment	Best For	Default Label
`itemize`	Unordered items	Bullets (•, –, *, ·)
`enumerate`	Ordered/sequential	Numbers (1., (a), i., A.)
`description`	Definitions/terms	Bold label text

Table 3.1: List environments summary

Chapter 4

Tables

4.1 Basic Table (tabular)

The `tabular` environment is the fundamental table environment in LaTeX.

4.1.1 Column Types

Spec	Description
`l`	Left-aligned column
`c`	Centered column
`r`	Right-aligned column
`p{width}`	Paragraph column with fixed width
`|`	Vertical line
`@{text}`	Replace inter-column space with *text*

Table 4.1: Column type specifications

4.1.2 Basic Example

```
\begin{tabular}{|l|c|r|}
    \hline
    Left & Center & Right \\
    \hline
    A    & B      & C \\
    \hline
\end{tabular}
```

Listing 4.1: Basic tabular

Result:

Left	Center	Right
A	B	C

4.2 Table Syntax

```latex
% &       -- column separator
% \\      -- end of row
% \hline -- horizontal line (spans full width)
% \cline{2-3} -- partial horizontal line (columns 2 to 3)

\begin{tabular}{|l|c|r|}
    \hline
    Name  & Age & Score \\
    \hline
    Alice & 25  & 95 \\
    Bob   & 30  & 87 \\
    \cline{2-3}
    \multicolumn{1}{c|}{} & \multicolumn{2}{c}{Partial line} \\
    \hline
\end{tabular}
```

Listing 4.2: Table syntax reference

4.3 Professional Tables with booktabs

The `booktabs` package produces publication-quality tables without vertical lines.

```latex
\usepackage{booktabs}

\begin{tabular}{lcc}
    \toprule
    Name  & Age & Score \\
    \midrule
    Alice & 25  & 95 \\
    Bob   & 30  & 87 \\
    Carol & 28  & 92 \\
    \bottomrule
\end{tabular}
```

Listing 4.3: booktabs example

Result:

Name	**Age**	**Score**
Alice	25	95
Bob	30	87
Carol	28	92
Average	**27.7**	**91.3**

Table 4.2: Student scores using booktabs

booktabs command	Purpose
`\toprule`	Top rule (thicker)
`\midrule`	Middle rule (medium)
`\bottomrule`	Bottom rule (thicker)
`\cmidrule{a-b}`	Partial rule (thinner)
`\addlinespace`	Extra vertical space

Table 4.3: booktabs commands

4.4 Multi-column and Multi-row

4.4.1 Multi-column (`\multicolumn`)

```
\multicolumn{number_of_columns}{alignment}{text}
```

Listing 4.4: Multicolumn

	Math	Physics	Chemistry
Alice	95	88	91
Bob	87	92	85
Class Average		**89.7**	

Table 4.4: Multicolumn example

4.4.2 Multi-row (multirow package)

```
\usepackage{multirow}
\multirow{number_of_rows}{width}{text}
```

Listing 4.5: Multirow

Category	Subject	Students	Avg Score
Science	Physics	45	88
	Chemistry	38	85
Arts	History	32	91
	Literature	29	93

Table 4.5: Multirow example

4.5 Fixed-width Tables (tabularx)

```
\usepackage{tabularx}

\begin{tabularx}{\textwidth}{lX}
```

```latex
    \toprule
    Name & Description \\
    \midrule
    Item 1 & This text automatically wraps to fill
             the available column width. \\
    Item 2 & Another long description that wraps
             automatically in the X column. \\
    \bottomrule
\end{tabularx}
```

Listing 4.6: tabularx

Result:

Command	Description
\LaTeX	Produces the LaTeX logo with proper kerning and raised "A".
\TeX	Produces the TeX logo.
\today	Produces the current date.
\tableofcontents	Generates the table of contents from sectioning commands.

Table 4.6: tabularx with auto-wrapping X column

4.6 Multi-page Tables (longtable)

```latex
\usepackage{longtable}

\begin{longtable}{lcc}
    \caption{A long table spanning multiple pages} \\
    \toprule
    Name & Value & Note \\
    \midrule
    \endfirsthead

    \toprule
    Name & Value & Note \\
    \midrule
    \endhead

    \midrule
    \multicolumn{3}{r}{\textit{Continued on next page}} \\
    \endfoot

    \bottomrule
    \endlastfoot

    Data1 & 100 & Note1 \\
    Data2 & 200 & Note2 \\
    % ... many more rows ...
```

```latex
\end{longtable}
```

Listing 4.7: longtable

4.7 Table Placement (Float)

Tables inside `table` environment are *floating bodies*:

```latex
\begin{table}[placement]
    % placement can be:
    % h = here        (approximately at current location)
    % t = top         (top of page)
    % b = bottom      (bottom of page)
    % p = page        (on a separate float page)
    % ! = override    (relax LaTeX's internal rules)
    % H = HERE        (requires float package -- exact placement)
    \centering
    \begin{tabular}{lc}
        ...
    \end{tabular}
    \caption{Table caption}
    \label{tab:mylabel}
\end{table}
```

Listing 4.8: Float placement specifiers

4.8 Colored Tables

```latex
\usepackage[table]{xcolor}

\rowcolor{gray!20}        % Color entire row
\cellcolor{blue!25}        % Color single cell
\columncolor{green!15}     % Color entire column (in column spec)
```

Listing 4.9: Colored table rows

Header 1	Header 2	Header 3
Row 1, Col 1	Row 1, Col 2	Row 1, Col 3
Row 2, Col 1	Row 2, Col 2	Row 2, Col 3
Row 3, Col 1	Row 3, Col 2	Row 3, Col 3
Row 4, Col 1	Row 4, Col 2	Row 4, Col 3

Table 4.7: Alternating row colors

4.9 Table Quick Reference

Symbol/Command	**Meaning**
`&`	Column separator
`\\`	Row terminator
`\hline`	Full horizontal line
`\cline{a-b}`	Partial horizontal line
`\vline`	Vertical line (inline)
`\multicolumn{n}{spec}{text}`	Span n columns
`\multirow{n}{w}{text}`	Span n rows
`\toprule`	booktabs top rule
`\midrule`	booktabs middle rule
`\bottomrule`	booktabs bottom rule

Table 4.8: Table syntax quick reference

Chapter 5

Mathematics

LaTeX is renowned for its mathematical typesetting. This chapter covers everything from basic inline formulas to advanced constructs.

5.1 Math Modes

There are several ways to enter math mode:

Type	Syntax	Usage
Inline math	`$...$`	$E = mc^2$ in text
Display math	`\[...\]`	Centered, numbered
Equation env	`\begin{equation}`	Centered, numbered
Align env	`\begin{align}`	Multi-line aligned

Table 5.1: Math mode entry points

```
% Inline: the formula flows with text
The energy is $E = mc^2$ according to Einstein.

% Display: the formula is centered on its own line
\[ E = mc^2 \]

% Equation: like display but with a number
\begin{equation}
    E = mc^2 \label{eq:einstein}
\end{equation}

% Starred version: no number
\begin{equation*}
    E = mc^2
\end{equation*}
```

Listing 5.1: Math modes

5.2 Superscripts and Subscripts

```
$x^2$                  % Superscript: x squared
$x_i$                  % Subscript: x sub i
$x_i^2$                % Both: x sub i squared
$x_{i,j}$              % Multi-character subscript: need braces
$a^{b^c}$              % Nested superscript
$2^{10} = 1024$        % Powers
```

Listing 5.2: Superscripts and subscripts

Results: x^2, x_i, x_i^2, $x_{i,j}$, a^{b^c}, $2^{10} = 1024$

5.3 Fractions

```
$\frac{a}{b}$                                % Basic fraction
$\dfrac{a}{b}$                               % Display-style fraction
$\tfrac{a}{b}$                               % Text-style fraction
$\cfrac{1}{1+\cfrac{1}{2}}$                 % Continued fraction
$\binom{n}{k}$                               % Binomial coefficient
```

Listing 5.3: Fractions

Results: $\frac{a}{b}$, $\dfrac{a}{b}$, $\cfrac{1}{1+\cfrac{1}{2}}$, $\binom{n}{k}$

5.4 Roots

```
$\sqrt{x}$              % Square root
$\sqrt[3]{x}$           % Cube root
$\sqrt[n]{x}$           % nth root
$\sqrt{a^2 + b^2}$      % Complex expression
```

Listing 5.4: Roots

Results: $\sqrt{x}$, $\sqrt[3]{x}$, $\sqrt[n]{x}$, $\sqrt{a^2 + b^2}$

5.5 Sums, Products, Integrals

```
$\sum_{i=1}^{n} x_i$          % Summation
$\prod_{i=1}^{n} x_i$         % Product
$\int_{a}^{b} f(x) dx$        % Definite integral
$\iint_{D} f(x,y) dA$         % Double integral
$\iiint_{V} f(x,y,z) dV$      % Triple integral
$\oint_{C} F \cdot dr$        % Contour integral
$\lim_{x \to \infty} f(x)$   % Limit
$\sup_{x \in S} f(x)$        % Supremum
$\max_{x \in S} f(x)$        % Maximum
```

```
$\min_{x \in S} f(x)$          % Minimum
```

Listing 5.5: Big operators

Display style results:

$$\sum_{i=1}^{n} x_i \qquad \prod_{i=1}^{n} x_i \qquad \int_a^b f(x)\,dx \qquad \lim_{x\to\infty} f(x)$$

5.6 Greek Letters

Command	Result	Command	Result
\alpha	α	\Alpha	A
\beta	β	\gamma	γ
\delta	δ	\epsilon	ϵ
\varepsilon	ε	\zeta	ζ
\eta	η	\theta	θ
\iota	ι	\kappa	κ
\lambda	λ	\mu	μ
\nu	ν	\xi	ξ
\pi	π	\rho	ρ
\sigma	σ	\tau	τ
\upsilon	υ	\phi	ϕ
\varphi	φ	\chi	χ
\psi	ψ	\omega	ω
\Gamma	Γ	\Delta	Δ
\Theta	Θ	\Lambda	Λ
\Xi	Ξ	\Pi	Π
\Sigma	Σ	\Phi	Φ
\Psi	Ψ	\Omega	Ω

Table 5.2: Greek letters

5.7 Mathematical Operators and Functions

```
$\sin x$        $\cos x$        $\tan x$
$\arcsin x$     $\arccos x$     $\arctan x$
$\sinh x$       $\cosh x$       $\tanh x$
$\ln x$         $\log x$        $\exp x$
$\det A$        $\dim V$        $\ker f$
$\deg f$        $\gcd(a,b)$     $\hom(A,B)$
$\inf S$        $\sup S$        $\max S$
$\min S$        $\lim f$        $\limsup f$
```

Listing 5.6: Math operators

5.8 Relations and Binary Operators

Command	Symbol	Command	Symbol
`<`	$<$	`>`	$>$
`\leq`	$\leq$	`\geq`	$\geq$
`\neq`	$\neq$	`\approx`	$\approx$
`\equiv`	$\equiv$	`\sim`	$\sim$
`\propto`	$\propto$	`\ll`	$\ll$
`\gg`	$\gg$	`\in`	$\in$
`\subset`	$\subset$	`\supset`	$\supset$
`\subseteq`	$\subseteq$	`\supseteq`	$\supseteq$
`\cup`	$\cup$	`\cap`	$\cap$
`\setminus`	$\setminus$	`\emptyset`	$\emptyset$
`\pm`	$\pm$	`\mp`	$\mp$
`\times`	$\times$	`\div`	$\div$
`\cdot`	$\cdot$	`\circ`	$\circ$
`\otimes`	$\otimes$	`\oplus`	$\oplus$

Table 5.3: Common mathematical symbols

5.9 Arrows

Command	Arrow	Command	Arrow
`\leftarrow`	$\leftarrow$	`\rightarrow`	$\rightarrow$
`\Leftarrow`	$\Leftarrow$	`\Rightarrow`	$\Rightarrow$
`\leftrightarrow`	$\leftrightarrow$	`\Leftrightarrow`	$\Leftrightarrow$
`\mapsto`	$\mapsto$	`\longmapsto`	$\longmapsto$
`\uparrow`	$\uparrow$	`\downarrow`	$\downarrow$
`\nearrow`	$\nearrow$	`\searrow`	$\searrow$

Table 5.4: Arrow symbols

5.10 Matrices

The `amsmath` package provides several matrix environments:

```latex
% Parentheses: ( )
\begin{pmatrix}
    a & b \\
    c & d
\end{pmatrix}

% Brackets: [ ]
\begin{bmatrix}
    a & b \\
```

```latex
    c & d
\end{bmatrix}

% Braces: { }
\begin{Bmatrix}
    a & b \\
    c & d
\end{Bmatrix}

% Vertical bars: | |
\begin{vmatrix}
    a & b \\
    c & d
\end{vmatrix}

% Double vertical bars: || ||
\begin{Vmatrix}
    a & b \\
    c & d
\end{Vmatrix}

% No delimiters
\begin{matrix}
    a & b \\
    c & d
\end{matrix}

% With dots:
\begin{pmatrix}
    a_{11} & a_{12} & \cdots & a_{1n} \\
    a_{21} & a_{22} & \cdots & a_{2n} \\
    \vdots & \vdots & \ddots & \vdots \\
    a_{m1} & a_{m2} & \cdots & a_{mn}
\end{pmatrix}
```

Listing 5.7: Matrices

Results:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

5.11 Multi-line Equations (align)

```latex
\begin{align}
    f(x) &= x^2 + 2x + 1 \\
```

```
            &= (x+1)^2 \\
            &= x^2 + 2x + 1 \label{eq:expand}
\end{align}

% align without numbering:
\begin{align*}
    f(x) &= x^2 + 2x + 1 \\
            &= (x+1)^2
\end{align*}

% Number specific lines only:
\begin{align}
    f(x) &= x^2 + 2x + 1 \nonumber \\
            &= (x+1)^2 \\
            &= x^2 + 2x + 1 \nonumber
\end{align}

% Multiple alignment columns:
\begin{align*}
    x &= 1  &  y &= 2  &  z &= 3 \\
    x' &= 4 &  y' &= 5 &  z' &= 6
\end{align*}
```

Listing 5.8: align environment

Result:

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x+1)^2 \end{aligned} \tag{5.1}$$

5.12 Cases and Piecewise Functions

```
\begin{equation}
    |x| =
    \begin{cases}
        x  & \text{if } x \geq 0 \\
        -x & \text{if } x < 0
    \end{cases}
\end{equation}
```

Listing 5.9: cases environment

Result:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \tag{5.2}$$

5.13 Accents in Math Mode

Command	Result	Name
`\hat{a}`	$\hat{a}$	Hat
`\bar{a}`	$\bar{a}$	Bar
`\tilde{a}`	$\tilde{a}$	Tilde
`\vec{a}`	$\vec{a}$	Vector arrow
`\dot{a}`	$\dot{a}$	Dot
`\ddot{a}`	$\ddot{a}$	Double dot
`\vec{a}`	$\vec{a}$	Vector
`\widehat{abc}`	$\widehat{abc}$	Wide hat
`\widetilde{abc}`	$\widetilde{abc}$	Wide tilde
`\overline{abc}`	$\overline{abc}$	Overline
`\underline{abc}`	$\underline{abc}$	Underline
`\overbrace{abc}`	$\overbrace{abc}$	Overbrace
`\underbrace{abc}`	$\underbrace{abc}$	Underbrace

Table 5.5: Math accents

5.14 Spacing in Math Mode

Command	Space	Example
`\,`	Thin space	$a\,b$
`\:`	Medium space	$a\:b$
`\;`	Thick space	$a\;b$
`\!`	Negative thin	$a\!b$
`\quad`	1em space	$a\quad b$
`\qquad`	2em space	$a\qquad b$
`\,`		ab (no space for comparison)

Table 5.6: Math spacing commands

5.15 Text Inside Math Mode

```
$\text{for all } x > 0$            % Regular text in math
$\text{if and only if}$             % Using amsmath \text
$\mathrm{d}x$                       % Upright d for differentials
$\text{Area} = \pi r^2$             % Upright labels
$\text{let } n = 5$                 % Mixed text and math
```

Listing 5.10: Text in math

5.16 Theorems and Proofs

```latex
% In preamble (amsthm package):
\newtheorem{theorem}{Theorem}[section]
\newtheorem{lemma}[theorem]{Lemma}
\newtheorem{corollary}[theorem]{Corollary}
\theoremstyle{definition}
\newtheorem{definition}{Definition}[section]
\theoremstyle{remark}
\newtheorem{remark}{Remark}

% In document:
\begin{theorem}[Pythagorean]
For a right triangle: $a^2 + b^2 = c^2$.
\end{theorem}

\begin{proof}
By construction of squares on each side...
\end{proof}
```

Listing 5.11: Theorem environments

Theorem 5.16.1 (Pythagorean)**.** *For a right triangle with legs a, b and hypotenuse c:*

$$a^2 + b^2 = c^2$$

Proof. Left as an exercise for the reader. □

Definition 5.16.1. A **prime number** is a natural number greater than 1 that has no positive divisors other than 1 and itself.

5.17 Comprehensive Math Example

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi} \tag{5.3}$$

$$e^{i\pi} + 1 = 0 \tag{5.4}$$

The Gaussian integral (5.3) and Euler's identity (5.4) are two of the most beautiful equations in mathematics.

Chapter 6

Figures & Images

6.1 Including Images (graphicx)

```latex
\usepackage{graphicx}

% Basic image:
\includegraphics{filename}

% With options:
\includegraphics[width=5cm]{filename}
\includegraphics[height=3cm]{filename}
\includegraphics[scale=0.5]{filename}
\includegraphics[width=\textwidth]{filename}
\includegraphics[width=0.8\textwidth]{filename}
\includegraphics[angle=90]{filename}        % Rotate 90 degrees
\includegraphics[clip=true, trim=1cm 2cm 1cm 2cm]{filename} % Crop

% Supported formats: PDF, PNG, JPG, EPS
```

Listing 6.1: Including images

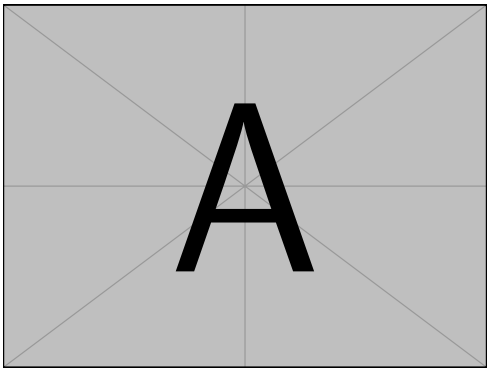


Figure 6.1: Example image A using `example-image-a`

6.2 Figure Environment (Float)

```latex
\begin{figure}[htbp]
    \centering
    \includegraphics[width=0.6\textwidth]{myimage}
    \caption{Description of the image.}
    \label{fig:mylabel}
\end{figure}

% Placement specifiers:
% h = here, t = top, b = bottom, p = float page
% ! = override LaTeX rules, H = exactly HERE (float package)
```

Listing 6.2: Figure float

6.2.1 Referencing Figures

As seen in Figure 6.1, we can reference images by their label.

```latex
% In text:
As shown in Figure~\ref{fig:mylabel} ...

% The ~ prevents a line break between "Figure" and the number.
```

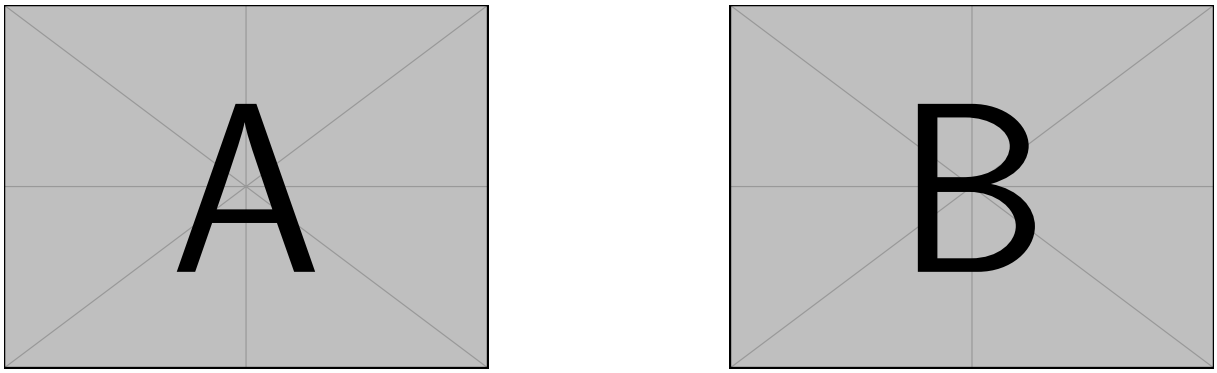
Listing 6.3: Referencing figures

6.3 Subfigures (subcaption package)

```latex
\usepackage{subcaption}

\begin{figure}[htbp]
    \centering
    \begin{subfigure}[b]{0.45\textwidth}
        \includegraphics[width=\textwidth]{image1}
        \caption{First subfigure}
        \label{fig:sub1}
    \end{subfigure}
    \hfill
    \begin{subfigure}[b]{0.45\textwidth}
        \includegraphics[width=\textwidth]{image2}
        \caption{Second subfigure}
        \label{fig:sub2}
    \end{subfigure}
    \caption{Overall caption for both subfigures}
    \label{fig:combined}
\end{figure}
```

Listing 6.4: Subfigures



(a) Subfigure A (b) Subfigure B

Figure 6.2: Two subfigures side by side

6.4 Image Size Options

Option	Description
`width=5cm`	Set exact width
`height=3cm`	Set exact height
`width=\textwidth`	Scale to text width
`width=0.5\textwidth`	Half text width
`scale=0.5`	Scale to 50%
`angle=45`	Rotate 45 degrees
`clip=true`	Clip to bounding box
`trim=l b r t`	Trim from left, bottom, right, top
`keepaspectratio`	Maintain proportions

Table 6.1: Image sizing options

6.5 Wrapping Text Around Images

```latex
\usepackage{wrapfig}

\begin{wrapfigure}{r}{0.4\textwidth}
    \centering
    \includegraphics[width=\linewidth]{image}
    \caption{Wrapped figure}
\end{wrapfigure}

This text will wrap around the figure on the left side.
% {r} = right side, {l} = left side, {R} = right (allows floating)
```

Listing 6.5: wrapfig

6.6 Drawing Simple Graphics with TikZ (Preview)

TikZ can also be used inside figure environments:

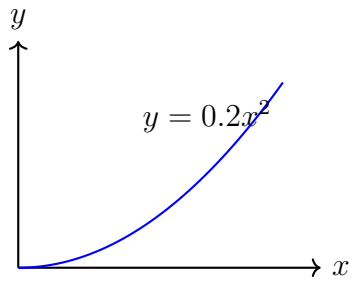


Figure 6.3: A simple plot drawn with TikZ

Chapter 7

Cross References & Hyperlinks

7.1 Labels and References

LaTeX provides a powerful cross-referencing system. Every numbered element can be labeled and referenced.

```latex
% Step 1: Place a label after the numbered element
\section{Introduction}
\label{sec:intro}

\begin{equation}
    E = mc^2
    \label{eq:einstein}
\end{equation}

\begin{figure}[htbp]
    \caption{My figure}
    \label{fig:myfig}
\end{figure}

\begin{table}[htbp]
    \caption{My table}
    \label{tab:mytab}
\end{table}

% Step 2: Reference it in text
As discussed in Section~\ref{sec:intro}...
Equation~\ref{eq:einstein} shows...
See Figure~\ref{fig:myfig} and Table~\ref{tab:mytab}.
```

Listing 7.1: Label and reference

7.1.1 Label Naming Conventions

Element	Prefix	Example
Chapter	`ch:`	`ch:intro`
Section	`sec:`	`sec:methods`
Subsection	`ssec:`	`ssec:data`
Figure	`fig:`	`fig:diagram`
Table	`tab:`	`tab:results`
Equation	`eq:`	`eq:euler`
Listing	`lst:`	`lst:python`
Item	`itm:`	`itm:first`

Table 7.1: Recommended label prefixes

7.1.2 Pageref and Nameref

```latex
% Reference the page number:
See page~\pageref{sec:intro}.

% Reference the title/name:
Section~\nameref{sec:intro} discusses...

% Reference both name and page with autoref (hyperref):
\autoref{sec:intro}  % Produces "Section 1"
\autoref{fig:myfig}  % Produces "Figure 1"
\autoref{eq:einstein}% Produces "Equation 1"
```

Listing 7.2: Other reference commands

7.2 Hyperlinks (hyperref package)

```latex
\usepackage{hyperref}

% URL links:
\url{https://www.latex-project.org/}
\href{https://www.latex-project.org/}{LaTeX Project}

% Email link:
\href{mailto:user@example.com}{user@example.com}

% File link:
\href{file:///path/to/file}{Open file}

% Internal link (jump to label):
\hyperref[sec:intro]{Go to Introduction}

% Configuration:
\hypersetup{
```

```
    colorlinks=true,        % Colored links (not boxed)
    linkcolor=blue,         % Internal link color
    filecolor=magenta,      % File link color
    urlcolor=cyan,          % URL color
    citecolor=green,        % Citation color
}
```

Listing 7.3: Hyperlinks

Examples:

- URL: https://www.latex-project.org/
- Hyperlinked text: Overleaf — Online LaTeX Editor

7.3 Table of Contents

```
% Basic table of contents:
\tableofcontents

% List of figures:
\listoffigures

% List of tables:
\listoftables

% Limit TOC depth:
\setcounter{tocdepth}{2}  % Show up to subsection

% Set sectioning number depth:
\setcounter{secnumdepth}{3}  % Number up to subsubsection

% Add unnumbered section to TOC:
\section*{Acknowledgments}
\addcontentsline{toc}{section}{Acknowledgments}
```

Listing 7.4: Table of contents

7.3.1 Custom TOC Entry

```
% Add arbitrary entries to TOC:
\addcontentsline{toc}{chapter}{Preface}

% For the list of figures/tables:
\addcontentsline{lof}{figure}{My Figure Title}
```

Listing 7.5: Custom TOC entries

Chapter 8

Bibliography & Citations

8.1 Basic Bibliography with thebibliography

The simplest way to create a bibliography is using the `thebibliography` environment directly.

```latex
% At the end of your document (before \end{document}):
\begin{thebibliography}{9}
    \bibitem{knuth1984}
        Donald E. Knuth,
        \textit{The TeXbook},
        Addison-Wesley, 1984.

    \bibitem{lamport1994}
        Leslie Lamport,
        \textit{LaTeX: A Document Preparation System},
        Addison-Wesley, 2nd edition, 1994.

    \bibitem{mittelbach2004}
        Frank Mittelbach and Michel Goossens,
        \textit{The LaTeX Companion},
        Addison-Wesley, 2nd edition, 2004.
\end{thebibliography}

% Cite in text:
Knuth wrote the original TeX system~\cite{knuth1984}.
LaTeX was created by Lamport~\cite{lamport1994}.
```

Listing 8.1: thebibliography environment

8.1.1 Citation Commands

Command	Output	Description
`\cite{key}`	[1]	Basic citation
`\cite[p.42]{key}`	[1, p.42]	With page number
`\cite{a,b,c}`	[1,2,3]	Multiple citations

Table 8.1: Basic citation commands

8.2 BibTeX with .bib Files

For larger projects, use a `.bib` file to manage references.

8.2.1 The .bib File

```
@article{einstein1905,
    author  = {Albert Einstein},
    title   = {On the Electrodynamics of Moving Bodies},
    journal = {Annalen der Physik},
    year    = {1905},
    volume  = {322},
    number  = {10},
    pages   = {891--921},
    doi     = {10.1002/andp.19053221004}
}

@book{knuth1984,
    author    = {Donald E. Knuth},
    title     = {The TeXbook},
    publisher = {Addison-Wesley},
    year      = {1984},
    isbn      = {978-0201134483}
}

@inproceedings{lamport1998,
    author    = {Leslie Lamport},
    title     = {The Part-Time Parliament},
    booktitle = {ACM Transactions on Computer Systems},
    year      = {1998},
    volume    = {16},
    number    = {2},
    pages     = {133--169}
}

@online{overleaf2024,
    author  = {{Overleaf}},
    title   = {Documentation},
    year    = {2024},
    url     = {https://www.overleaf.com/learn}
```

```
}

@phdthesis{smith2020,
    author = {Jane Smith},
    title  = {Advanced Topics in Computer Science},
    school = {MIT},
    year   = {2020}
}
```

Listing 8.2: Sample .bib file (references.bib)

8.2.2 Entry Types

Type	Used For
`@article`	Journal articles
`@book`	Books
`@booklet`	Booklets / pamphlets
`@inbook`	A chapter or section in a book
`@incollection`	A chapter in a collected works
`@inproceedings`	Conference papers
`@manual`	Technical documentation
`@mastersthesis`	Master's thesis
`@misc`	Miscellaneous
`@phdthesis`	PhD thesis
`@proceedings`	Conference proceedings
`@techreport`	Technical reports
`@unpublished`	Unpublished works
`@online`	Web resources

Table 8.2: BibTeX entry types

8.2.3 Common Fields

Field	Required	Description
`author`	Most entries	Author names
`title`	Most entries	Title of work
`year`	Most entries	Year of publication
`publisher`	book, inbook	Publisher name
`journal`	article	Journal name
`volume`	article	Volume number
`number`	article	Issue number
`pages`	article	Page range
`doi`	Optional	Digital Object Identifier
`url`	Optional	URL to resource
`isbn`	Optional	ISBN number
`abstract`	Optional	Abstract text
`keywords`	Optional	Keywords

Table 8.3: Common BibTeX fields

8.3 Using BibTeX

```latex
% In your .tex file, at the end:
\bibliographystyle{plain}     % Choose a style
\bibliography{references}     % Name of .bib file (without .bib)

% Compilation steps:
% 1. pdflatex main.tex       (first pass)
% 2. bibtex main              (process citations)
% 3. pdflatex main.tex       (second pass)
% 4. pdflatex main.tex       (third pass for references)
```

Listing 8.3: Using BibTeX

8.3.1 Bibliography Styles

Style	Description
`plain`	Alphabetical by author, numbered [1]
`unsrt`	Same as plain, but in order of citation
`alpha`	Labels like [Knu84] (author-year abbrev.)
`abbrv`	Like plain, with abbreviated first names
`ieeetr`	IEEE Transactions style
`acm`	ACM style
`apalike`	APA-like style (author, year)

Table 8.4: Common bibliography styles

8.4 Biblatex (Modern Approach)

```latex
% In preamble:
\usepackage[backend=biber, style=numeric]{biblatex}
\addbibresource{references.bib}

% In document body:
As shown by~\textcite{einstein1905}...
According to~\autocite{knuth1984}...

% At end of document:
\printbibliography

% Or with a title:
\printbibliography[title={Works Cited}]

% Or filter by type:
\printbibliography[type=article, title={Journal Articles}]
\printbibliography[type=book, title={Books}]

% Citation commands unique to biblatex:
\cite{key}            % [1]
\parencite{key}       % (Author, Year)
\textcite{key}        % Author (Year)
\autocite{key}        % Depends on style
\footcite{key}        % Footnote citation
```

Listing 8.4: biblatex package

8.5 Example Bibliography

Here is a working example with inline references:

Bibliography

[1] Donald E. Knuth, *The TEXbook*, Addison-Wesley, 1984.

[2] Leslie Lamport, *LATEX: A Document Preparation System*, Addison-Wesley, 2nd edition, 1994.

[3] Frank Mittelbach and Michel Goossens, *The LATEX Companion*, Addison-Wesley, 2nd edition, 2004.

[4] Overleaf, *Learn LATEX in 30 Minutes*, https://www.overleaf.com/learn, 2024.

Chapter 9

Code Listings & Verbatim Text

9.1 Verbatim Text

The `verbatim` environment prints text exactly as typed, without interpreting any LaTeX commands.

```latex
% Inline verbatim:
\verb|This is \verbatim{test} text|

% Display verbatim:
\begin{verbatim}
    \textbf{This text is NOT bold}
    $E = mc^2$ is NOT an equation
    Everything appears exactly as typed!
\end{verbatim}

% Verbatim with different delimiter:
\verb!Special chars: $ % & # _ { }!

% With fancyvrb package (more options):
\usepackage{fancyvrb}
\begin{Verbatim}[fontsize=\small, frame=single, framesep=2mm]
    Custom verbatim with frame and smaller font
\end{Verbatim}
```

Listing 9.1: Verbatim environments

9.1.1 Inline Verbatim Examples

- `\documentclass{article}` — the document class command
- `\usepackage{graphicx}` — loading a package
- `$E = mc^2$` — inline math

9.2 Code Listings (listings package)

The `listings` package provides sophisticated source code formatting.

9.2.1 Basic Listing

```python
def hello_world():
    print("Hello, World!")
```

Listing 9.2: Basic code listing

9.2.2 Listing with Language

```python
# Python example: Fibonacci sequence
def fibonacci(n):
    """Calculate the nth Fibonacci number."""
    if n <= 1:
        return n
    a, b = 0, 1
    for _ in range(2, n + 1):
        a, b = b, a + b
    return b

# Generate first 10 Fibonacci numbers
for i in range(10):
    print(f"F({i}) = {fibonacci(i)}")
```

Listing 9.3: Python example with syntax highlighting

9.2.3 Java Example

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");

        // Loop example
        for (int i = 0; i < 5; i++) {
            System.out.println("Iteration: " + i);
        }
    }
}
```

Listing 9.4: Java example

9.2.4 C/C++ Example

```cpp
#include <iostream>
#include <vector>
#include <algorithm>

```

```cpp
int main() {
    std::vector<int> numbers = {5, 2, 8, 1, 9, 3};

    // Sort ascending
    std::sort(numbers.begin(), numbers.end());

    // Print sorted vector
    for (const auto& n : numbers) {
        std::cout << n << " ";
    }
    std::cout << std::endl;

    return 0;
}
```

Listing 9.5: C++ example

9.2.5 JavaScript Example

```javascript
// Asynchronous data fetching
async function fetchData(url) {
    try {
        const response = await fetch(url);
        const data = await response.json();
        return data;
    } catch (error) {
        console.error("Error:", error);
    }
}
```

Listing 9.6: JavaScript example

9.3 Listing Styles and Configuration

```latex
% Global settings in preamble:
\lstset{
    basicstyle=\ttfamily\small,            % Base font
    keywordstyle=\color{blue}\bfseries,    % Keywords
    commentstyle=\color{green!50!black}\itshape, % Comments
    stringstyle=\color{red!70!black},      % Strings
    numbers=left,                          % Line numbers
    numberstyle=\tiny\color{gray},         % Line number style
    stepnumber=1,                          % Line number step
    numbersep=5pt,                         % Distance of numbers
    backgroundcolor=\color{gray!5},        % Background color
    frame=single,                          % Border frame
    rulecolor=\color{gray!50},             % Frame color
    breaklines=true,                       % Auto line break
    breakatwhitespace=false,               % Break only at whitespace
    tabsize=4,                             % Tab size
```

```
    showstringspaces=false,                % Don't show string spaces
    captionpos=b,                          % Caption at bottom
    title=\lstname,                        % Show filename as title
}
```

Listing 9.7: Listing configuration

Note: Per-listing options go in square brackets: `\begin{lstlisting}[language=Python, caption={...}]`.

9.4 Inline Code

```
% Use \lstinline for inline code:
The function \lstinline|print("hello")| outputs text.

% Short form (define a shorthand):
\lstMakeShortInline[language=Python]{|}
Use |print("hello")| in your code.
```

Listing 9.8: Inline code

9.5 Supported Languages

The listings package supports many programming languages:

Language	Keyword	Language	Keyword
Python	`Python`	Java	`Java`
C	`C`	C++	`C++`
JavaScript	`Java`	HTML	`HTML`
SQL	`SQL`	R	`R`
MATLAB	`Matlab`	Fortran	`Fortran`
Bash	`bash`	XML	`XML`
LaTeX	`[LaTeX]TeX`	Make	`make`

Table 9.1: Some supported languages in the listings package

Chapter 10

TikZ Graphics

TikZ ("TikZ ist kein Zeichenprogramm") is a powerful package for creating graphics directly in LaTeX.

10.1 Basic Shapes

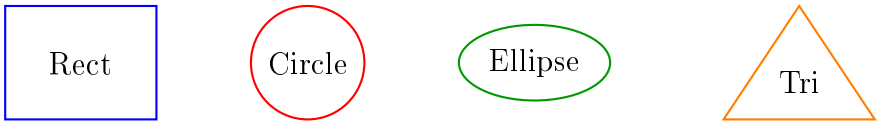


Figure 10.1: Basic shapes in TikZ

```
\begin{tikzpicture}
    \draw (0,0) rectangle (2,1);          % Rectangle
    \draw (4,1) circle (1);                % Circle
    \draw (7,1) ellipse (1.5 and 0.5);    % Ellipse
    \draw (0,0) -- (2,0) -- (1,2) -- cycle;  % Triangle
\end{tikzpicture}
```

Listing 10.1: Basic shapes

10.2 Lines and Paths

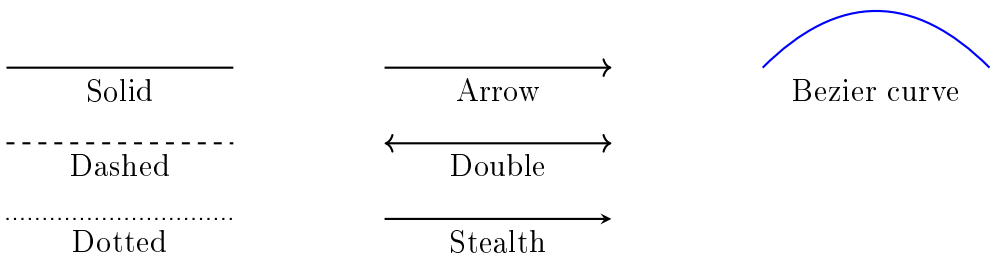


Figure 10.2: Lines and paths in TikZ

10.3 Filling and Colors

Figure 10.3: Filling and colors in TikZ

10.4 Nodes and Labels

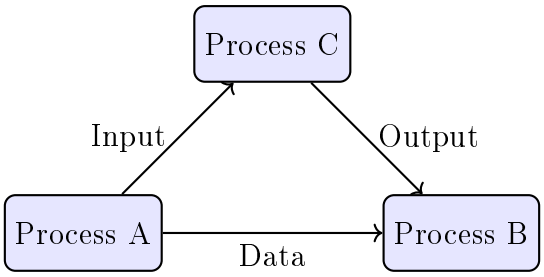


Figure 10.4: Nodes and connections in TikZ

```
\begin{tikzpicture}[
    box/.style={rectangle, draw, thick, fill=blue!10,
                minimum width=2cm, minimum height=1cm}
]
    \node[box] (a) at (0,0) {Node A};
    \node[box] (b) at (5,0) {Node B};

    \draw[->, thick] (a) -- (b) node[midway, above] {Label};
    \draw[->, thick, bend left=30] (a) to (b);
    \draw[->, thick, bend right=30] (a) to[bend right] (b);
\end{tikzpicture}
```

Listing 10.2: Nodes and edges

10.5 Flowchart

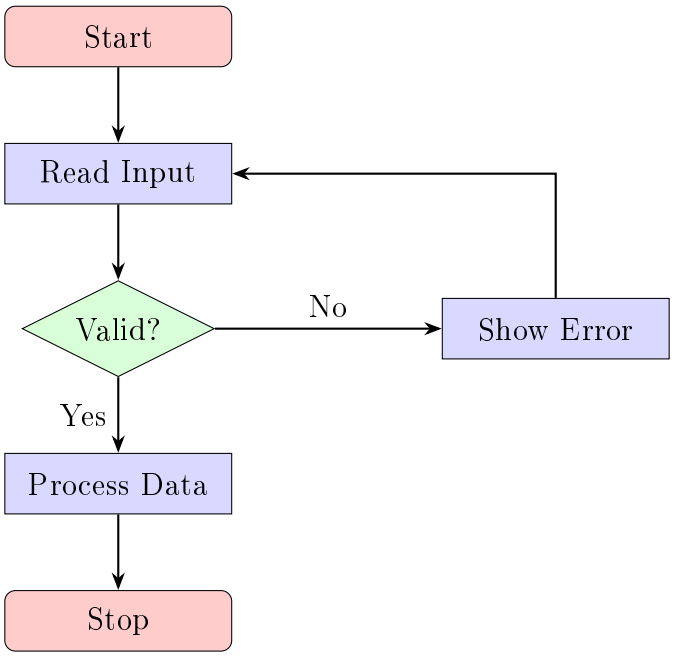


Figure 10.5: Flowchart using TikZ (requires `shapes.geometric` and `positioning` libraries)

10.6 Plots with pgfplots

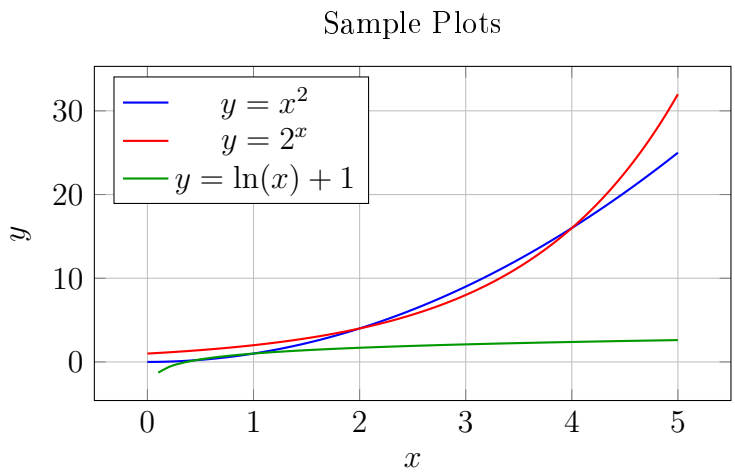


Figure 10.6: Function plots using pgfplots

```
\usepackage{pgfplots}
\pgfplotsset{compat=1.18}

\begin{tikzpicture}
    \begin{axis}[
        xlabel={$x$}, ylabel={$y$},
```

```
        legend pos=north west,
        grid=major,
    ]
        \addplot[blue, thick, domain=0:5, samples=50] {x^2};
        \addlegendentry{$y = x^2$}

        \addplot[red, thick, domain=0:5] {2^x};
        \addlegendentry{$y = 2^x$}
    \end{axis}
\end{tikzpicture}
```

Listing 10.3: pgfplots example

10.7 Bar Chart

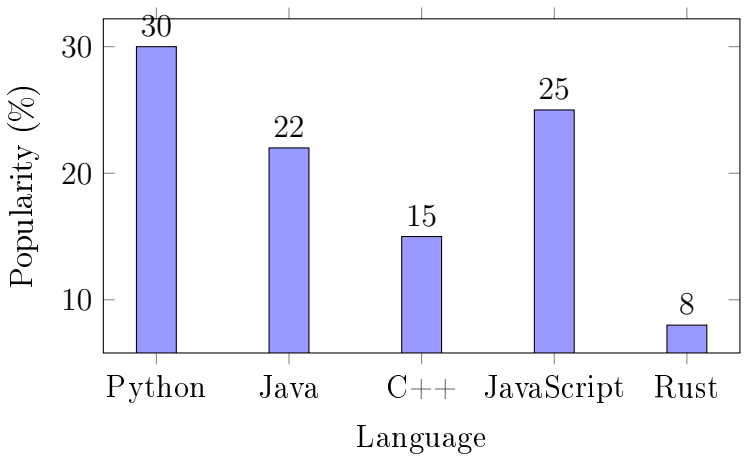


Figure 10.7: Bar chart using pgfplots

10.8 TikZ Styles

```
% Define reusable styles:
\tikzset{
    mybox/.style={
        rectangle, draw, thick,
        fill=blue!10,
        minimum width=2cm,
        minimum height=1cm,
        rounded corners=3pt,
        font=\sffamily
    },
    myarrow/.style={
        ->, >=stealth, thick, shorten >=1pt
    },
}

% Use them:
```

```
\node[mybox] (a) at (0,0) {Styled};
\node[mybox, fill=red!10] (b) at (3,0) {Override};
\draw[myarrow] (a) -- (b);
```

Listing 10.4: TikZ styles

Chapter 11

Page Layout & Headers/Footers

11.1 Page Geometry (geometry package)

```latex
\usepackage{geometry}

% Set all margins at once:
\geometry{margin=2.5cm}

% Set individual margins:
\geometry{
    left=3cm,
    right=3cm,
    top=2cm,
    bottom=2cm,
    bindingoffset=5mm   % Extra for binding
}

% Landscape orientation:
\geometry{landscape}

% Paper size:
\geometry{a4paper}     % or letterpaper, legalpaper, a5paper
```

Listing 11.1: Page layout

11.1.1 Understanding Page Layout

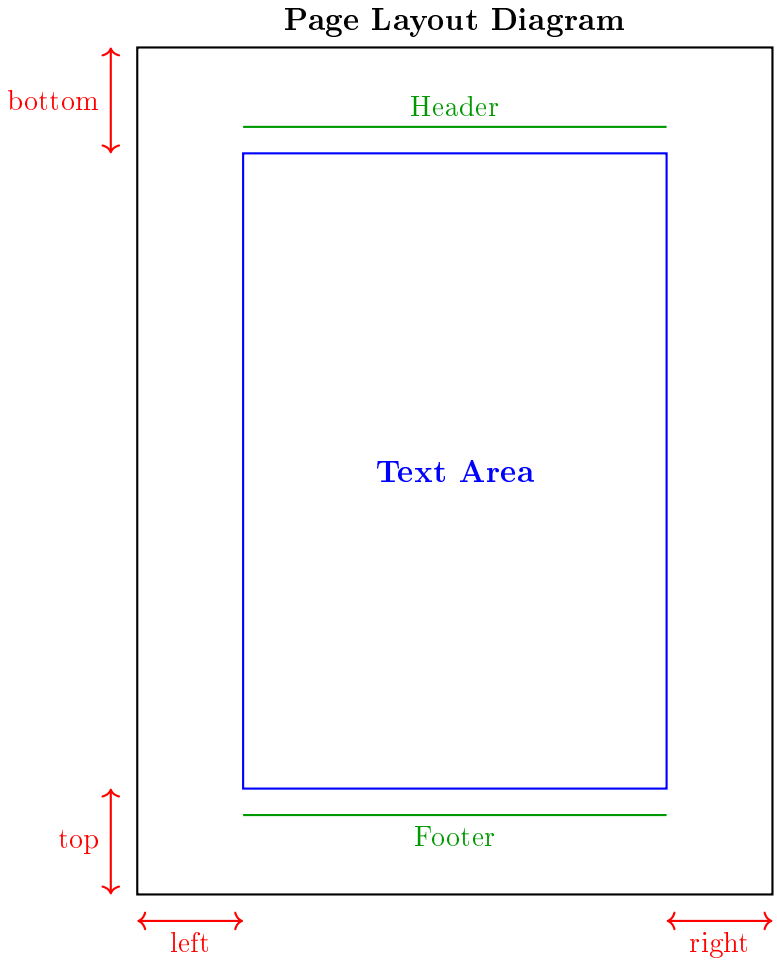


Figure 11.1: Page layout anatomy

11.2 Headers and Footers (fancyhdr)

```
\usepackage{fancyhdr}
\pagestyle{fancy}

% Clear all defaults:
\fancyhf{}

% Set header:
\fancyhead[L]{Left header}        % L = Left
\fancyhead[C]{Center header}      % C = Center
\fancyhead[R]{Right header}       % R = Right

% Set footer:
\fancyfoot[L]{Author Name}
\fancyfoot[C]{\thepage}           % Page number
\fancyfoot[R]{\today}

% For even/odd pages (twoside):
```

```latex
\fancyhead[LE,RO]{\thepage}       % Left Even, Right Odd
\fancyhead[RE,LO]{\leftmark}      % Right Even, Left Odd

% Rule thickness:
\renewcommand{\headrulewidth}{0.4pt}
\renewcommand{\footrulewidth}{0pt}  % No footer rule

% Chapter page style (plain):
\fancypagestyle{plain}{
    \fancyhf{}
    \fancyfoot[C]{\thepage}
    \renewcommand{\headrulewidth}{0pt}
}
```

Listing 11.2: fancyhdr package

11.2.1 Position Codes

Code	Position
L	Left
C	Center
R	Right
E	Even page
O	Odd page
LE	Left on even page
RO	Right on odd page

Table 11.1: Header/footer position codes

11.3 Page Numbering

```latex
% Numbering styles:
\pagenumbering{arabic}    % 1, 2, 3 (default)
\pagenumbering{roman}     % i, ii, iii
\pagenumbering{Roman}     % I, II, III
\pagenumbering{alph}      % a, b, c
\pagenumbering{Alph}      % A, B, C

% Set page counter:
\setcounter{page}{1}      % Start counting from 1

% No page number on current page:
\thispagestyle{empty}
```

Listing 11.3: Page numbering

11.4 Multiple Columns

```latex
% Document-wide twocolumn option:
\documentclass[twocolumn]{article}

% Switch mid-document:
\onecolumn    % Switch to one column
\twocolumn    % Switch to two columns

% With multicol package (more flexible):
\usepackage{multicol}

\begin{multicols}{3}  % 3 columns
    Text in column one continues into
    column two and then column three.
\end{multicols}

\begin{multicols}{2}[Section Title]
    Two-column section with a title.
\end{multicols}
```

Listing 11.4: Multiple columns

11.5 Page Breaks

```latex
\newpage          % Start a new page
\clearpage        % Flush all floats, then new page
\cleardoublepage  % For twoside: go to next odd-numbered page
\pagebreak        % Suggest a page break (stretchable)
\nopagebreak      % Suggest no page break
\linebreak        % Suggest a line break
\nolinebreak      % Suggest no line break
```

Listing 11.5: Page breaks

11.6 Paragraph Control

```latex
% Paragraph indent:
\setlength{\parindent}{1cm}          % Set indent width
\noindent                            % No indent for this paragraph
\indent                              % Force indent

% Paragraph skip (no indent, vertical space instead):
\usepackage{parskip}                % Removes indent, adds space

% Or manually:
\setlength{\parindent}{0pt}
\setlength{\parskip}{6pt plus 2pt minus 1pt}
```

Listing 11.6: Paragraph control

Chapter 12

Floating Bodies & Captions

12.1 What Are Floats?

Floats are objects (figures, tables) that LaTeX places at optimal positions in the document, rather than exactly where they appear in the source code.

Environment	Content
`figure`	Images, diagrams
`table`	Tables
`algorithm`	Algorithms (with algorithm package)
`listing`	Code listings (with listings package)

Table 12.1: Float environments

12.2 Placement Specifiers

Specifier	Meaning	Notes
`h`	Here	Approximately at the current location
`t`	Top	At the top of the page
`b`	Bottom	At the bottom of the page
`p`	Page	On a separate float page
`!`	Override	Override LaTeX's internal placement rules
`H`	HERE	Exactly here (requires `float` package)

Table 12.2: Float placement specifiers

```latex
\begin{figure}[htbp]      % Try here, top, bottom, then page
    ...
\end{figure}

\begin{figure}[H]         % Exactly here (float package)
    ...
```

```latex
\end{figure}

\begin{table}[!ht]        % Override rules, try here then top
    ...
\end{table}
```

Listing 12.1: Float placement

12.3 Captions

```latex
% Caption below the content (typical for figures):
\begin{figure}[htbp]
    \centering
    \includegraphics[width=5cm]{image}
    \caption{This is a figure caption.}
    \label{fig:myfigure}
\end{figure}

% Caption above the content (typical for tables):
\begin{table}[htbp]
    \centering
    \caption{This is a table caption.}
    \label{tab:mytable}
    \begin{tabular}{lc}
        ...
    \end{tabular}
\end{table}
```

Listing 12.2: Captions

12.3.1 Caption Styles (caption package)

```latex
\usepackage{caption}

% Change label separator:
\captionsetup{labelsep=period}   % Figure 1. Caption text
\captionsetup{labelsep=quad}     % Figure 1    Caption text
\captionsetup{labelsep=newline}  % Label on separate line
\captionsetup{labelsep=none}     % No label

% Font options:
\captionsetup{font=small}
\captionsetup{font=it}
\captionsetup{labelfont=bf}

% Justification:
\captionsetup{justification=centering}
\captionsetup{justification=raggedright}
\captionsetup{justification=justified}

```

```
% Per-environment:
\captionsetup[figure]{font=small, labelfont=bf}
\captionsetup[table]{font=small, skip=5pt}
```

Listing 12.3: Caption customization

12.4 Side Captions

```
\usepackage{sidecap}

\begin{SCfigure}
    \includegraphics[width=5cm]{image}
    \caption{Caption appears beside the figure.}
\end{SCfigure}
```

Listing 12.4: Side captions

12.5 Float Barriers

Sometimes floats drift too far from their reference. Use barriers to control this:

```
\usepackage{placeins}

% Force all pending floats to be placed before this point:
\FloatBarrier

% Use in a section command:
\makeatletter
\AtBeginSection{\FloatBarrier}
\makeatother
```

Listing 12.5: Float barriers

12.6 Controlling Float Counters

```
% Maximum floats per page:
\renewcommand{\topfraction}{0.9}      % Max fraction of page for top floats
\renewcommand{\bottomfraction}{0.9}   % Max fraction for bottom floats
\renewcommand{\textfraction}{0.05}    % Min fraction for text
\renewcommand{\floatpagefraction}{0.8} % Min fraction of floats for float page

% Maximum number of floats:
\setcounter{topnumber}{4}             % Max top floats per page
\setcounter{bottomnumber}{4}          % Max bottom floats per page
\setcounter{totalnumber}{10}          % Max total floats per page
```

Listing 12.6: Float counters

Chapter 13

Essential Packages & Their Uses

13.1 Package Management

```latex
% Basic syntax:
\usepackage{packagename}

% With options:
\usepackage[option1, option2]{packagename}

% Multiple packages at once:
\usepackage{package1, package2, package3}
```

Listing 13.1: Loading packages

Note: Packages must be loaded in the preamble (before `\begin{document}`*). Load* `hyperref` *last (or near-last) as a general rule.*

13.2 Essential Package Reference

Package	Category	Purpose
`inputenc`	Encoding	UTF-8 input encoding
`fontenc`	Encoding	Font encoding (T1)
`geometry`	Layout	Page margins and layout
`graphicx`	Graphics	Include images
`amsmath`	Math	Advanced math typesetting
`amssymb`	Math	Additional math symbols
`amsthm`	Math	Theorem environments
`booktabs`	Tables	Professional table rules
`array`	Tables	Extended column types
`tabularx`	Tables	Flexible-width tables
`longtable`	Tables	Multi-page tables
`multirow`	Tables	Multi-row cells
`hyperref`	Links	Hyperlinks and cross-refs
`xcolor`	Color	Color support

Package	Category	Purpose
`listings`	Code	Source code formatting
`tikz`	Graphics	Programmable drawings
`pgfplots`	Graphics	Plots and charts
`fancyhdr`	Layout	Headers and footers
`caption`	Floats	Caption customization
`subcaption`	Floats	Subfigures/subtables
`enumitem`	Lists	Custom list formatting
`parskip`	Layout	Paragraph spacing
`float`	Floats	Improved float placement
`siunitx`	Science	SI units and numbers
`url`	Links	URL line breaking
`fancyvrb`	Verbatim	Enhanced verbatim
`multicol`	Layout	Multiple columns
`wrapfig`	Floats	Text-wrapped figures
`natbib`	Bibliography	Flexible citations
`biblatex`	Bibliography	Modern bibliography
`glossaries`	Reference	Glossary / acronyms
`index`	Reference	Index generation
`nomencl`	Reference	Nomenclature lists
`appendix`	Structure	Appendix formatting
`titlesec`	Structure	Section title formatting
`tocloft`	TOC	TOC customization
`setspace`	Layout	Line spacing control
`ulem`	Text	Underline / strikethrough
`textcomp`	Symbols	Additional text symbols
`placeins`	Floats	Float barriers

13.3 Package Conflicts and Loading Order

```latex
% 1. Encoding
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}

% 2. Language (babel or polyglossia)
\usepackage[english]{babel}

% 3. Fonts
\usepackage{lmodern}   % Latin Modern fonts

% 4. Math
\usepackage{amsmath, amssymb, amsthm}

% 5. Graphics and drawing
\usepackage{graphicx}
\usepackage{tikz}

% 6. Tables
```

```latex
\usepackage{booktabs, array, tabularx}

% 7. Layout
\usepackage{geometry}
\usepackage{fancyhdr}

% 8. Lists and text
\usepackage{enumitem}
\usepackage{xcolor}

% 9. Code
\usepackage{listings}

% 10. Bibliography
\usepackage{natbib}  % or biblatex

% 11. Hyperref (ALWAYS LAST or near-last)
\usepackage{hyperref}

% 12. Appendix (after hyperref)
\usepackage{appendix}
```

Listing 13.2: Recommended loading order

13.4 SI Units (siunitx)

```latex
\usepackage{siunitx}

% Number formatting:
\num{12345.67}                % 12 345.67
\num{1.23e4}                  % 1.23 x 10^4
\num[round-precision=2]{3.14159}  % 3.14

% Units:
\si{kg}                       % kg
\si{m/s}                      % m/s
\si{kg.m.s^{-2}}              % kg m s^-2
\si{\meter\per\second}        % m/s (symbolic)

% Value + unit:
\SI{9.8}{m/s^2}               % 9.8 m/s^2
\SI{100}{\degreeCelsius}      % 100 degrees C
\SI{3.14}{\micro\meter}       % 3.14 micrometer
```

Listing 13.3: siunitx package

Examples: The speed of light is 3×10^8 m/s. Water boils at 100 °C. The wavelength is 550 nm.

13.5 Creating Your Own Package

```latex
% Save as mystyle.sty:
\NeedsTeXFormat{LaTeX2e}
\ProvidesPackage{mystyle}[2024/01/01 My Custom Style]

% Package options:
\DeclareOption{draft}{\typeout{Draft mode}}
\ProcessOptions\relax

% Load required packages:
\RequirePackage{amsmath}
\RequirePackage{graphicx}

% Define commands:
\newcommand{\emphasis}[1]{\textbf{\textit{#1}}}

\endinput
% Then in your document:
% \usepackage{mystyle}
```

Listing 13.4: Creating a package

Chapter 14

Custom Commands & Environments

14.1 New Commands (\newcommand)

```latex
% Basic: no arguments
\newcommand{\myname}{John Doe}
Hello, I am \myname.          % Output: Hello, I am John Doe.

% With arguments (1 required argument):
\newcommand{\greet}[1]{Hello, #1!}
\greet{Alice}                 % Output: Hello, Alice!

% With multiple arguments:
\newcommand{\add}[2]{#1 + #2}
$\add{a}{b}$                  % Output: a + b

% With optional arguments:
\newcommand{\hello}[2][World]{Hello, #2 from #1!}
\hello{Alice}                 % Output: Hello, Alice from World!
\hello[Bob]{Alice}            % Output: Hello, Alice from Bob!

% Maximum 9 arguments (#1 through #9)
```

Listing 14.1: newcommand syntax

14.1.1 Practical Examples

```latex
% Shorthand for common formatting:
\newcommand{\pkg}[1]{\texttt{#1}}
\newcommand{\cmd}[1]{\texttt{\textbackslash#1}}

% Math shortcuts:
\newcommand{\R}{\mathbb{R}}                % Real numbers
\newcommand{\norm}[1]{\left\|#1\right\|} % Norm notation
\newcommand{\abs}[1]{\left|#1\right|}     % Absolute value
\newcommand{\inner}[2]{\langle #1, #2 \rangle} % Inner product
\newcommand{\dd}{\,\mathrm{d}}            % Differential d

```

```latex
% Usage:
% $f\colon \R^n \to \R$
% $\norm{x} = \sqrt{\inner{x}{x}}$
% $\int_0^1 f(x) \dd x$
```

Listing 14.2: Practical custom commands

Results:

$$f\colon \mathbb{R}^n \to \mathbb{R}, \quad \|x\| = \sqrt{\langle x, x\rangle}$$

14.2 Renaming Commands (\renewcommand)

```latex
% Override existing commands:
\renewcommand{\contentsname}{Contents}          % Rename TOC title
\renewcommand{\figurename}{Fig.}                % Rename figure label
\renewcommand{\tablename}{Table}                % Rename table label
\renewcommand{\bibname}{References}             % Rename bibliography

% Check if command exists:
\providecommand{\mycommand}{default value}    % Define only if not
    already
```

Listing 14.3: renewcommand

14.3 New Environments

You can define entire environments (begin/end pairs) with `\newenvironment`:

The syntax is: `\newenvironment{name}[args]{begin-code}{end-code}`

Example 1 – a simple warning box:

```latex
\newenvironment{warning}
    {\begin{center}\bfseries Warning: }
    {\end{center}}
```

Example 2 – a box with a title argument:

```latex
\newenvironment{infobox}[1]
    {\begin{center}\fbox{\parbox{0.9\textwidth}%
     {\textbf{##1}\par\smallskip}}}
    {\end{center}}
```

14.4 Lengths and Counters

14.4.1 Defining Lengths

Custom lengths

```latex
% Create a new length:
\newlength{\mywidth}
```

```
\setlength{\mywidth}{5cm}

% Use it:
\hspace{\mywidth}

% Arithmetic on lengths:
\addtolength{\mywidth}{2cm}           % Add 2cm
\settowidth{\mywidth}{Hello World}    % Set to width of text
\settoheight{\myheight}{Text}         % Set to height
\settodepth{\mydepth}{Text}           % Set to depth
```

14.4.2 Standard Lengths

Length	Controls
\textwidth	Width of text area
\linewidth	Width of current line
\columnwidth	Width of column
\paperwidth	Width of paper
\paperheight	Height of paper
\parindent	Paragraph indentation
\parskip	Space between paragraphs
\baselineskip	Distance between baselines
\columnsep	Space between columns

Table 14.1: Standard LaTeX lengths

14.4.3 Counters

```
% Define a new counter:
\newcounter{mycounter}

% Set, step, and use:
\setcounter{mycounter}{0}
\stepcounter{mycounter}       % Increment by 1
\addtocounter{mycounter}{5}   % Add 5

% Display:
\arabic{mycounter}    % 1, 2, 3
\roman{mycounter}     % i, ii, iii
\Roman{mycounter}     % I, II, III
\alph{mycounter}      % a, b, c
\Alph{mycounter}      % A, B, C
\fnsymbol{mycounter} % *, dagger, double-dagger, etc.
```

Listing 14.4: Counters

14.5 Conditional Commands

```latex
% ifthen package:
\usepackage{ifthen}

\newcommand{\checkodd}[1]{%
    \ifthenelse{\isodd{#1}}%
        {#1 is odd}%
        {#1 is even}%
}

% Boolean flags:
\newboolean{draftmode}
\setboolean{draftmode}{true}

\ifthenelse{\boolean{draftmode}}%
    {\watermark{DRAFT}}%
    {}
```

Listing 14.5: Conditionals

14.6 Boxes

Boxes

```
% mbox: prevents line breaks
\mbox{This text stays on one line}

% fbox: framed box
\fbox{Framed text}

% makebox: custom box with alignment
\makebox[5cm][c]{Centered in 5cm}
% Alignment: l=left, c=center, r=right, s=stretch

% parbox: paragraph in a box
\parbox{3cm}{This text wraps inside a 3cm-wide box.}

% minipage: like parbox but more powerful
\begin{minipage}[t]{0.45\textwidth}
    Left column content.
\end{minipage}
\hfill
\begin{minipage}[t]{0.45\textwidth}
    Right column content.
\end{minipage}

% raisebox: vertically offset text
\raisebox{5pt}{Raised text}
\raisebox{-5pt}{Lowered text}
```

Side-by-side minipage example:

Left Column
This content sits in the left minipage. It wraps automatically within the specified width.

Right Column
This content sits in the right minipage. Using minipages we can create multi-column layouts anywhere.

Chapter 15

Advanced Topics

15.1 Tables of Contents for Non-Standard Sections

```
% Add unnumbered chapter to TOC:
\chapter*{Preface}
\addcontentsline{toc}{chapter}{Preface}

% Custom TOC depth:
\setcounter{tocdepth}{2}      % Show sections and subsections only
\setcounter{secnumdepth}{3}   % Number up to subsubsection

% Mini table of contents per chapter (minitoc package):
\usepackage{minitoc}
\dominitoc                        % In preamble, after \
    tableofcontents
\chapter{My Chapter}
\minitoc                          % Shows TOC for this chapter only
```

Listing 15.1: Custom TOC entries

15.2 Index

```
\usepackage{makeidx}
\makeindex                         % In preamble

% Mark entries:
\index{keyword}
\index{main topic!subtopic}       % Nested index entry
\index{term|textbf}               % Bold page reference

% Special characters:
\index{keyword@\texttt{keyword}}  % Custom sort vs display

% Print the index:
\printindex                        % At end of document

```

```
% Compilation:
% pdflatex main.tex
% makeindex main
% pdflatex main.tex
```

Listing 15.2: Index

15.3 Glossary / Acronyms

```
\usepackage[acronym, toc]{glossaries}
\makeglossaries

% Define terms:
\newglossaryentry{latex}{
    name=LaTeX,
    description={A document preparation system}
}

\newacronym{api}{API}{Application Programming Interface}

% Use in text:
\gls{latex}          % First: "LaTeX (A document preparation system)"
                      % Later: "LaTeX"
\gls{api}            % First: "Application Programming Interface (API)"
                      % Later: "API"
\Gls{latex}          % Capitalized first letter

% Print glossary:
\printglossary[type=main]           % Glossary
\printglossary[type=\acronymtype]   % Acronyms
```

Listing 15.3: Glossary

15.4 Watermarks

```
\usepackage[firstpage]{draftwatermark}
\SetWatermarkText{CONFIDENTIAL}
\SetWatermarkScale{3}
\SetWatermarkColor[gray]{0.9}
\SetWatermarkAngle{45}

% Or with wallpaper package for images:
\usepackage{wallpaper}
\ThisCenterWallPaper{0.5}{logo.png}
```

Listing 15.4: Watermark

15.5 Conditional Compilation

```latex
% Define a flag:
\newif\ifdraft
\drafttrue            % Set to draft mode
% \draftfalse         % Set to final mode

% Use conditionally:
\ifdraft
    \usepackage[draft]{graphicx}  % Show image boxes only
\else
    \usepackage{graphicx}         % Full images
\fi

% Include content conditionally:
\ifdraft
    \include{chapters/draft-notes}
\fi

% Version control:
\newcommand{\version}{1.0}
\newcommand{\releasedate}{2024-01-01}
```

Listing 15.5: Conditional compilation

15.6 External Files and Shell Escape

```latex
% Read from external file:
\input{data/variables.tex}        % Insert file content

% Write to file (during compilation):
\newwrite\myoutput
\immediate\openout\myoutput=output.txt
\immediate\write\myoutput{Hello from LaTeX}
\immediate\closeout\myoutput

% Shell escape (compile with --shell-escape):
% \write18{date > date.txt}
% \input{date.txt}
```

Listing 15.6: File operations

15.7 Compilation Tools

15.7.1 latexmk

```bash
# latexmk automates multi-pass compilation:
latexmk -pdf main.tex              % Compile to PDF
latexmk -xelatex main.tex          % Use XeLaTeX
latexmk -lualatex main.tex         % Use LuaLaTeX
```

```
latexmk -c                          % Clean auxiliary files
latexmk -C                          % Clean all (including PDF)
latexmk -pvc main.tex               % Preview continuously (live
   reload)
```

Listing 15.7: latexmk automation

15.7.2 Compilation Engines

Engine	Command	Best For
pdfLaTeX	`pdflatex`	Standard documents, English
XeLaTeX	`xelatex`	Unicode, system fonts
LuaLaTeX	`lualatex`	Advanced scripting, Lua integration

Table 15.1: LaTeX compilation engines

15.7.3 Using System Fonts (XeLaTeX/LuaLaTeX)

```
% Requires XeLaTeX or LuaLaTeX
\usepackage{fontspec}

% Set main font:
\setmainfont{Times New Roman}
\setsansfont{Arial}
\setmonofont{Consolas}

% With options:
\setmainfont[
    BoldFont=Times New Roman Bold,
    ItalicFont=Times New Roman Italic,
    Scale=1.0
]{Times New Roman}
```

Listing 15.8: System fonts with fontspec

15.8 Beamer Presentations (Overview)

```
\documentclass{beamer}
\usetheme{Madrid}
\usecolortheme{default}

\title{My Presentation}
\author{Author Name}
\date{\today}

\begin{document}

\begin{frame}
```

```latex
    \titlepage
\end{frame}

\begin{frame}{Outline}
    \tableofcontents
\end{frame}

\begin{frame}{My First Slide}
    \begin{itemize}
        \item First point
        \item Second point
    \end{itemize}
\end{frame}

% Columns:
\begin{frame}{Two Columns}
    \begin{columns}
        \column{0.5\textwidth}
            Left content
        \column{0.5\textwidth}
            Right content
    \end{columns}
\end{frame}

% Overlays (step-by-step reveal):
\begin{frame}{Overlays}
    \begin{itemize}
        \item<1-> First (appears on slide 1+)
        \item<2-> Second (appears on slide 2+)
        \item<3-> Third (appears on slide 3+)
    \end{itemize}
\end{frame}
\end{document}
```

Listing 15.9: Beamer basics

15.9 Multilingual Support

```latex
% With babel:
\usepackage[english, french, german]{babel}
% Last language is default. Switch:
\selectlanguage{french}
\selectlanguage{english}

% With polyglossia (XeLaTeX/LuaLaTeX):
\usepackage{polyglossia}
\setdefaultlanguage{english}
\setotherlanguage{french}
\setotherlanguage{german}

\begin{german}
```

```
    Hallo Welt!
\end{german}
```

Listing 15.10: Multilingual documents

15.10 Error Handling

Error	Common Cause
`Undefined control sequence`	Misspelled command or missing package
`Missing $ inserted`	Used math symbol outside math mode
`Extra alignment tab`	Too many `&` in table row
`Missing } inserted`	Unmatched braces
`Float too large`	Figure/table wider than text area
`Overfull \hbox`	Text extends past margin
`Underfull \hbox`	Too much whitespace in line

Table 15.2: Common LaTeX errors

Note: Always read the `.log` file for detailed error information. The first error reported is usually the root cause.

Appendix A

Quick Reference Card

A.1 Document Skeleton

```latex
\documentclass[12pt, a4paper]{report}

% === PREAMBLE ===
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{geometry}
\usepackage{graphicx}
\usepackage{amsmath, amssymb}
\usepackage{booktabs}
\usepackage{hyperref}
\usepackage{xcolor}
\usepackage{listings}
\usepackage{tikz}

\title{My Document Title}
\author{Author Name}
\date{\today}

% === BODY ===
\begin{document}
\maketitle
\tableofcontents

\chapter{Introduction}
\label{ch:intro}

\section{Background}
\label{sec:background}

Text with \textbf{bold}, \textit{italic}, and \texttt{monospace}.

Inline math: $E = mc^2$

Display math:
```

```latex
\[ \int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \]

\begin{figure}[htbp]
    \centering
    \includegraphics[width=0.6\textwidth]{myimage}
    \caption{My figure.}
    \label{fig:myfigure}
\end{figure}

\begin{table}[htbp]
    \centering
    \caption{My table.}
    \label{tab:mytable}
    \begin{tabular}{lcc}
        \toprule
        Name & Value & Note \\
        \midrule
        A    & 1     & x \\
        B    & 2     & y \\
        \bottomrule
    \end{tabular}
\end{table}

\bibliographystyle{plain}
\bibliography{references}

\end{document}
```

Listing A.1: Complete document skeleton

A.2 Symbol Quick Reference

Symbol	Code	Symbol	Code
α	`\alpha`	β	`\beta`
γ	`\gamma`	δ	`\delta`
ϵ	`\epsilon`	θ	`\theta`
λ	`\lambda`	μ	`\mu`
π	`\pi`	σ	`\sigma`
ϕ	`\phi`	ω	`\omega`
$\sum$	`\sum`	$\prod$	`\prod`
$\int$	`\int`	$\sqrt{x}$	`\sqrt{x}`
$\frac{a}{b}$	`\frac{a}{b}`	∞	`\infty`
$\forall$	`\forall`	$\exists$	`\exists`
$\in$	`\in`	$\notin$	`\notin`
$\subset$	`\subset`	$\cup$	`\cup`
$\cap$	`\cap`	$\emptyset$	`\emptyset`
$\to$	`\to`	$\Rightarrow$	`\Rightarrow`
$\leq$	`\leq`	$\geq$	`\geq`
$\neq$	`\neq`	$\approx$	`\approx`
$\pm$	`\pm`	$\times$	`\times`
$\cdot$	`\cdot`	$\ldots$	`\ldots`
$\cdots$	`\cdots`	$\vdots$	`\vdots`

Table A.1: Common math symbols quick reference

A.3 Useful Commands Cheat Sheet

Command	Purpose
`\documentclass{article}`	Set document type
`\usepackage{pkg}`	Load a package
`\begin{env}...\end{env}`	Use an environment
`\textbf{text}`	Bold text
`\textit{text}`	Italic text
`\texttt{text}`	Monospace text
`\underline{text}`	Underlined text
`\textcolor{red}{text}`	Colored text
`\footnote{text}`	Footnote
`\cite{key}`	Citation
`\ref{label}`	Cross reference
`\pageref{label}`	Page reference
`\label{key}`	Define label
`\includegraphics{file}`	Include image
`\tableofcontents`	Table of contents
`\listoffigures`	List of figures
`\listoftables`	List of tables
`\appendix`	Start appendix
`\today`	Current date
`\LaTeX`	LaTeX logo
`\TeX`	TeX logo

Table A.2: Essential commands cheat sheet

A.4 Compilation Workflow

Task	Commands
Basic compile	`pdflatex main.tex`
With BibTeX	`pdflatex main` → `bibtex main` → `pdflatex main` ×2
With glossary	Add `makeglossaries main` after first compile
Automated	`latexmk -pdf main.tex`
Clean aux	`latexmk -c`

Table A.3: Compilation workflows